

Assignment # 1
Deep Learning
AI and MLP Framework for Osteoporosis Detection Using Population Health Data
Submission Deadline: 22 Feb 2026

1. Osteoporosis and Its Global Burden

Osteoporosis is a chronic metabolic bone disorder characterized by reduced bone mineral density and deterioration of bone structure, leading to weakened bones and increased fracture risk; healthy bone has dense trabeculae and a thick cortical layer, whereas osteoporosis causes trabecular thinning, loss of connectivity, and cortical porosity, making bones fragile even during routine activities. Globally, osteoporosis affects over 200 million people and is a growing public health concern due to aging populations, with one in three women and one in five men over 50 years experiencing an osteoporotic fracture; more than 8.9 million fractures occur annually, equivalent to one fracture every three seconds, and postmenopausal women and older adults are most vulnerable due to hormonal changes, reduced peak bone mass, and age related bone loss. Fractures of the spine, hip, and wrist result in substantial morbidity and mortality, with hip fractures showing one year mortality rates of 20 to 30 percent in older adults, while vertebral fractures are the most common but remain undiagnosed in nearly two thirds of cases due to silent or mild symptoms, increasing future fracture risk and long term disability. The World Health Organization defines osteoporosis as bone mineral density 2.5 standard deviations below the young adult mean, with osteopenia as an intermediate stage; however, despite clear criteria, osteoporosis remains underdetected because dual energy X ray absorptiometry, the diagnostic gold standard, is used by fewer than 30 percent of high risk individuals worldwide due to high cost, limited availability, and need for trained personnel, especially in resource limited settings, leading to diagnosis after fractures and irreversible damage. In Pakistan, the burden is aggravated by widespread vitamin D deficiency, low calcium intake, sedentary lifestyle, limited sunlight exposure, poor awareness, and limited access to preventive healthcare, emphasizing the need for accessible, low cost, and scalable early screening methods.

2. Data Source and Availability

Data for this assignment is obtained from the National Health and Nutrition Examination Survey (NHANES), a publicly available dataset maintained by the Centers for Disease Control and Prevention (CDC) that provides nationally representative health data for the non institutionalized civilian population of the United States using a multistage probability sampling design. NHANES integrates interviews, standardized physical examinations, laboratory based analyses, and imaging measurements, enabling comprehensive assessment of demographic, anthropometric, biochemical, lifestyle, and clinical factors relevant to multifactorial conditions such as osteoporosis. Survey cycles from 1999 to 2018 were combined to form a large and diverse adult cohort, increasing statistical power and generalizability. The dataset includes age, sex, height, weight, body mass index, serum vitamin D, calcium, alkaline phosphatase, and smoking history. Spine bone mineral density measured by dual energy X ray absorptiometry (DXA) was used only to define Normal, Osteopenia, and Osteoporosis, while all predictive features were limited to non invasive routine variables. Download data using following link

https://docs.google.com/spreadsheets/d/1jPKzuA5tZ7a1NFikNb0E8WoyU_feoPG/edit?usp=sharing&ouid=107556213524139231559&rtpof=true&sd=true

3. Selected Features and Their Relationship with Osteoporosis

Osteoporosis is a multifactorial condition influenced by demographic, anthropometric, biochemical, hormonal, and lifestyle factors. The features selected for analysis were chosen based on clinical relevance, availability in routine healthcare settings, and evidence from prior medical literature. These variables capture both direct and indirect pathways through which bone health is affected. The table below summarizes the key features used and explains their relationship with osteoporosis:

Feature	Category	Relationship with Osteoporosis
Age	Demographic	Bone mass declines with advancing age due to reduced bone formation and increased resorption, making age one of the strongest predictors of osteoporosis.
Gender	Demographic	Women are at higher risk due to lower peak bone mass and accelerated bone loss after menopause.
Body Mass Index (BMI)	Anthropometric	Low BMI is associated with reduced mechanical loading on bones and lower estrogen levels, both of which contribute to bone loss.
Height	Anthropometric	Taller individuals may experience greater vertebral load; height loss can also indicate vertebral compression fractures.
Weight	Anthropometric	Lower body weight is linked to reduced bone density, while moderate weight can have a protective mechanical effect.
Serum Vitamin D	Biochemical	Vitamin D deficiency impairs calcium absorption and bone mineralization, increasing fracture risk.
Serum Calcium	Biochemical	Calcium is essential for bone structure; chronic low levels contribute to reduced bone mineral density.
Alkaline Phosphatase (ALP)	Biochemical	Elevated ALP reflects increased bone turnover, commonly seen in osteoporosis.
Hemoglobin (Hb)	Biochemical	Low hemoglobin may reflect poor nutritional status or chronic disease, indirectly affecting bone health.
Hematocrit (HCT)	Biochemical	Reduced hematocrit is associated with frailty and metabolic imbalance, which may correlate with bone loss.
Menopause Status	Hormonal	Estrogen deficiency after menopause accelerates bone resorption, leading to rapid loss of bone density.
Smoking Status	Lifestyle	Smoking negatively affects osteoblast function and calcium absorption, increasing osteoporosis risk.
Diabetes History	Clinical	Diabetes is associated with altered bone quality and higher fracture risk despite normal or near-normal BMD.

Assignment Tasks:

Task 1: Data Visualization

Visualize the following features using histograms, scatter plots, and box plots to understand their distribution and relationships with the target variable:

- **Age**
- **BMI**
- **Serum Vitamin D**
- **Serum Calcium**

Plot Types:

- **Histogram:** For understanding feature distributions. Also understand class-wise distribution
- **Scatter Plot:** To visualize correlations between features.
- **Box Plot:** To observe the spread and outliers in features.

Task 2: MLP Implementation

Develop and train a Multilayer Perceptron (MLP) neural network using the provided data. The following steps should be followed:

1. **Network Architecture:**
 - Implement light to deep network and check BIAS and VARIANCE of network and comment on those
 - Use different activation functions, learning rates and optimization techniques
 - Apply cross-entropy as the loss function, and accuracy as the evaluation metric.
2. **Training and Evaluation:**
 - Divide data in 80-20 split as training and testing. Sub divide training data for validation as well
 - Train the MLP on the training data.
 - Evaluate the model on the testing set and record the performance.

Task 3: Ablation Studies

Perform ablation studies by varying the following parameters. Document and analyse the effect of each change on the model's performance:

1. **Changing the Optimizer:**
 - Compare Adam and SGD optimizers and record the changes in accuracy and loss.

2. Changing the Activation Function:

- Compare the use of ReLU, Leaky ReLU, Sigmoid, and Tanh activation functions. Report the results.

3. Changing the Number of Neurons and Layers:

- Experiment with varying the number of neurons and layers to check BIAS and VARIANCE

4. Changing the Learning Rate:

- Experiment with different learning rates such as 0.01, 0.001, and 0.0001. Compare their effects on the model's performance.

Do Ablation studies by keeping initially given parameters and then changing one at a time.

Deliverable:

Make a detailed report and add all results as graphs, bar graphs, tables etc